

Azure Databricks | Data Engineering Cheat Sheet

One-page reference - key concepts with ready-to-say sentences & architecture overview

CORE DATA ENGINEERING CONCEPTS

- Lakehouse Architecture**
A lakehouse combines the low-cost scalability of a data lake with the structured query and ACID guarantees of a data warehouse. It lets you store raw files cheaply while still running SQL analytics directly on them.
- Medallion Architecture (Bronze / Silver / Gold)**
Data flows through three layers: Bronze stores raw ingested data as-is, Silver is cleaned and deduplicated, and Gold holds business-level aggregates ready for dashboards and reporting.
- Delta Lake**
Delta Lake is an open-source storage layer that adds ACID transactions, schema enforcement, and time-travel to Parquet files. It is the default table format in Databricks.
- Unity Catalog**
Unity Catalog provides centralized governance across all Databricks workspaces, managing permissions, data lineage, and audit logging from a single control plane.
- Apache Spark (Distributed Compute)**
Spark is the distributed engine under Databricks that parallelizes data processing across a cluster. You write transformations in Python/SQL and Spark handles partitioning and execution.
- ETL / ELT Pipelines**
ETL extracts data, transforms it, then loads it into a target. Modern cloud pipelines often use ELT: load raw data first, then transform in-place using Spark or SQL.
- Auto Loader & Structured Streaming**
Auto Loader incrementally ingests new files as they land in cloud storage. Structured Streaming extends this to near-real-time, processing micro-batches continuously.
- Delta Live Tables (DLT)**
DLT is a declarative ETL framework: you define what each table should look like and Databricks handles orchestration, retries, and data quality checks automatically.
- Schema Evolution**
Schema evolution lets pipelines adapt when source data adds or changes columns, without breaking existing downstream tables or queries.
- Data Quality / Expectations**
Expectations are constraints applied during transformation, e.g. 'temperature must be between -50 and 80'. Rows that fail can be dropped, flagged, or quarantined.

KEY AZURE + TERRAFORM RESOURCES

- azurerm_databricks_workspace**
The workspace is the primary Databricks resource in Azure. It provisions the managed control plane, VNet injection, and the web UI endpoint.
- azurerm_storage_account + containers**
ADLS Gen2 storage accounts host the data lake. Containers map to medallion layers (bronze, silver, gold) and are accessed via abfss:// URIs.
- azurerm_databricks_access_connector**
A managed identity that lets Databricks authenticate to Azure storage without storing credentials. Required for Unity Catalog external locations.
- databricks_catalog + schema**
The catalog is the top-level namespace in Unity Catalog. Schemas (databases) sit inside catalogs and contain tables, views, and volumes.
- databricks_storage_credential + external_location**
Storage credentials link an Azure managed identity to Unity Catalog. External locations map specific ADLS paths so they can be governed by the catalog.
- databricks_cluster + cluster_policy**
Clusters are the compute. Policies set guardrails (max nodes, allowed instance types, auto-termination) to control cost and compliance.
- databricks_job**
Jobs orchestrate notebooks or tasks on a schedule or trigger. They support multi-task DAGs with retries and alerting.
- azurerm_key_vault + secrets**
Key Vault stores sensitive values (tokens, connection strings). Databricks secret scopes can back onto Key Vault for secure access from notebooks.
- azurerm_role_assignment**
RBAC bindings grant identities (e.g. the access connector) roles like Storage Blob Data Contributor on storage accounts.
- databricks_grants**
Fine-grained Unity Catalog permissions: grant SELECT, USE CATALOG, CREATE TABLE etc. to users or groups on catalogs, schemas, and tables.

ARCHITECTURE OVERVIEW

